

Reflective Lantern - Innovation Report

New Project Built

Name: Traffic-Pulse

Branch: <https://github.com/atharvadevne123/reflective-lantern/tree/innovation/traffic-pulse>

Tag: traffic-pulse-v1.0.0 Date: 2026-07-08

Inspired by: Urban mobility / smart-city ML portfolio gap (topic chosen via duplicate-gate elimination)

What it does: Real-time urban traffic congestion prediction and incident detection API using XGBoost-LightGBM ensemble with temporal features, route scoring, and automated drift detection.

NOTE: Session GitHub access was scoped to reflective-lantern, so the standalone Traffic-Pulse repo could not be created. Run the "Publish Traffic-Pulse Repo" workflow from the Actions tab (with a repo-scope PAT) for one-click publishing.

Tech Stack Coverage: 100% (10/10 required)

- [X] Python 3.11
- [X] FastAPI (7 endpoints)
- [X] XGBoost + LightGBM ensemble
- [X] sklearn Pipeline (26 features)
- [X] Model monitoring (KS-drift + DB logging)
- [X] Docker + docker-compose + PostgreSQL
- [X] SQLAlchemy + Alembic
- [X] pytest suite (85 tests)
- [X] .env.example
- [X] GitHub Actions CI (ruff + pytest)

Optional stack items implemented (7/4 required)

- [X] Time-series lag/rolling features
- [X] Drift detection (KS-test)
- [X] Automated retraining pipeline
- [X] 5-fold CV + AUC-ROC
- [X] Rate limiting middleware
- [X] Correlation-ID logging
- [X] API versioning /api/v1

Project Stats

Files created: 40 Tests: 85 Commits: 60

Ensemble AUC (5-fold CV, weighted OvR): 0.9700

xgb: AUC 0.9697 +/- 0.0028

lgbm: AUC 0.9702 +/- 0.0027